

Simulating Institutional Dynamics: A Multi-Agent Framework for Predicting Legal Reform Outcomes Using Extended Phenotype Theory

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Abstract

Jensen Huang observed in March 2026 that biology is transitioning from science to engineering through computation. This paper argues that the same transition applies to law. Legal systems, like biological organisms, exhibit phenotypic expression (institutional dynamics) driven by an underlying genotype (constitutional structure). Drawing on Extended Phenotype Theory (EPT) as formalized for legal-institutional dynamics in prior work (Lerer, 2026a), this paper treats legal norms as cultural replicators whose fitness determines survival in an institutional ecosystem.

Building on the Constitutional Lock-in Index (CLI) developed in prior work (Lerer, 2026a), I validate this four-component metric of structural institutional rigidity against four jurisdictions: Argentina (0.89), Spain (0.51), Brazil (0.40), and Chile (0.24). I implement a multi-agent simulation engine adapting the OASIS framework (Yang et al., 2024) for institutional dynamics, with seven agent types, nine legal actions, and two novel theoretical mechanisms: Heteronomous Bayesian Updating (HBU) and the Coefficient of Resistance to Institutional Change (CRI). Monte Carlo validation against 23 Argentine labour reforms (1974-2024) produces emergent CLI of 0.92 (target 0.89, error 0.033) with 93.8% reform-blocking in adversarial scenarios. Leave-one-out cross-validation across 60 cases in four countries yields 56.7% accuracy (95% CI: 44.1-68.4%), above a 43.3% baseline. I submit the programme to Mario Bunge's demarcation criteria and identify both strengths and unresolved weaknesses, particularly regarding the measurability of utilities in game-theoretic components.

Keywords: extended phenotype theory, evolutionary game theory, institutional dynamics, legal simulation, multi-agent systems, constitutional lock-in, regulatory hysteresis, synthetic populations

JEL Classification: K00, D02, C63, O17

1. Introduction

1.1 From Science to Engineering

In March 2026, Jensen Huang, CEO of NVIDIA, made a claim about biology that carries unintended implications for law. He argued that biology is transitioning from a science, characterized by sporadic discovery through trial and error, to an engineering discipline where each contribution compounds on previous ones and improvement becomes exponential. The mechanism he described is straightforward: computational simulation replaces physical experimentation. Millions of simulated protein folds at near-zero marginal cost, each one training the foundational model, before a single molecule is synthesized in a laboratory.

The legal parallel is precise. Traditional norm-making operates through a process remarkably similar to pre-computational drug discovery: a government proposes a reform, the legislature debates it, the executive enacts it, and then society waits years, sometimes decades, to discover whether the reform survived judicial review, institutional resistance, and political reversal. The 23 Argentine labour reforms documented in the companion dataset span fifty years (1974-2024); most were blocked, reversed, or rendered ineffective. The cost of this institutional trial-and-error is measured not in failed molecules but in failed policies, wasted political capital, and populations trapped in suboptimal institutional equilibria.

This paper proposes to do for legal reform prediction what computational biology did for protein folding: replace artisanal experimentation with systematic simulation. The tools now exist. In February 2026, Shapira et al. demonstrated that institutional simulation using synthetic populations of intelligent agents produces validated, reproducible results. The OASIS framework (Yang et al., 2024) supports up to one million interacting agents with individual personalities, long-term memory, and behavioural logic. What was missing was the adaptation of these tools to legal-institutional dynamics: the right agents, the right actions, the right metrics. That is what this paper provides.

1.2 The Demarcation Question

Applying biological terminology to law carries an obvious epistemic risk: that it amounts to metaphor dressed as theory. Mario Bunge, the Argentine physicist and philosopher who spent decades at McGill University developing criteria to distinguish genuine science from pseudoscience, would have asked the right question: does this programme generate falsifiable predictions about observable entities, or does it merely redecorate familiar phenomena with evolutionary vocabulary?

I take this question seriously throughout the paper. Bunge's demarcation criteria require, among other things, that a scientific field operate on real entities, use empirical methods, generate predictions that data can refute, and that research results modify the field's own components.

Section 4 tests these requirements against the simulation results. Where the programme succeeds, I report the data. Where it falls short, particularly regarding the measurability of utility functions in the game-theoretic components, I document the gap honestly. Bunge considered several applications of game theory pseudoscientific precisely because utilities are subjective and manipulable. The partial response offered here, observable fitness metrics based on adoption and litigation rates, does not fully resolve this objection. It is a limitation, not a concession.

1.3 Contribution and Road Map

This paper makes four contributions. First, building on the Extended Phenotype Theory framework formalized for legal-institutional dynamics in Lerer (2026a), I implement a multi-agent simulation engine that adapts OASIS for institutional dynamics, incorporating two novel mechanisms: Heteronomous Bayesian Updating (HBU) and the Coefficient of Resistance to Institutional Change (CRI). Second, I demonstrate that the Constitutional Lock-in Index (CLI), developed in prior work, emerges from agent interactions without target injection, validating it as a structural property rather than a descriptive label. Third, I validate the engine against historical data using Monte Carlo simulation and leave-one-out cross-validation across 60 cases in four jurisdictions. Fourth, I provide a public replication package that allows independent verification of all results reported here.

Section 2 develops the theoretical framework. Section 3 describes the simulation architecture. Section 4 presents empirical validation. Section 5 discusses limitations and implications. Section 6 concludes.

2. Theoretical Framework

2.1 Extended Phenotype Theory Applied to Law

Richard Dawkins introduced the concept of the extended phenotype in 1982 to argue that the gene's phenotypic effects extend beyond the organism's body. A beaver's dam is a phenotypic expression of beaver genes, just as beaver fur is. The dam modifies the environment in ways that affect the gene's fitness, creating a feedback loop between genotype, phenotype, and selection environment.

I apply this framework to legal systems. The constitutional structure of a country, its amendment rules, its distribution of veto powers, its judicial review mechanisms, functions as a genotype. The institutional dynamics that emerge from this structure, patterns of reform, resistance, capture, and reversal, are phenotypic expressions. Legal norms, in this framework, are cultural replicators

(memes) that compete for adoption, mutate through judicial interpretation, and survive or go extinct based on their fitness in the institutional environment.

This is not mere analogy. The claim is functional: the same mathematical machinery that describes gene-frequency change in biological populations (replicator dynamics) can describe norm-frequency change in institutional populations. The fitness of a legal norm can be operationalized as the ratio of its adoption rate to its litigation rate. A norm that is widely adopted and rarely litigated has high fitness; one that is frequently challenged in court has low fitness, regardless of its formal validity. This operationalization addresses one of Bunge's central objections: that utilities in social-science models are subjective. Adoption rates and litigation rates are observable, countable quantities.

2.2 The Constitutional Lock-in Index (CLI)

The CLI measures the structural rigidity of an institutional system: its resistance to reform, independent of the ideological direction of the reform. High CLI means that reforms in any direction are difficult; low CLI means the system can adapt in multiple directions. The formula is:

$$CLI = 0.30 \times \text{VetoPlayers} + 0.25 \times \text{JudicialReview} + 0.25 \times \text{AmendmentDifficulty} + 0.20 \times \text{InformalNorms}$$

Each component is operationalized as follows. VetoPlayers counts the number of institutional actors with effective blocking power, normalized to [0, 1]. JudicialReview measures the intensity of constitutional review, combining the frequency and success rate of constitutional challenges. AmendmentDifficulty captures the procedural barriers to formal constitutional change. InformalNorms measures the gap between formal rules and actual behaviour, using the ratio of norm evasion to norm compliance observed in the system.

Table 1: CLI Calibration Across Four Jurisdictions

Country	CLI	VetoPlayers	JudicialReview	AmendDifficulty	InformalNorms
Argentina	0.89	0.95	0.90	0.85	0.80
Spain	0.51	0.55	0.50	0.55	0.40
Brazil	0.40	0.45	0.40	0.40	0.30
Chile	0.24	0.25	0.20	0.30	0.20

Three clarifications are necessary. First, the weights (0.30, 0.25, 0.25, 0.20) were fixed before validation and have not been adjusted retroactively to improve fit. This is a commitment against ad hoc elaboration, one of Bunge's requirements for non-pseudoscientific practice. Second, the CLI is orthogonal to democracy indices. Argentina scores high on both Polity IV (democratic) and CLI (rigid). Chile scores high on Polity IV and low on CLI. The two measure different things: Polity IV

measures regime type; CLI measures structural reform capacity. A democracy can be locked in. Third, the component values in Table 1 are theoretically motivated estimates, not laboratory measurements. I make no stronger claim.

2.3 Evolutionary Game Theory Components

Three additional constructs complete the theoretical framework.

2.3.1 Institutional Hysteresis Rate (IHR)

The IHR measures the degree to which institutional deformation caused by a temporary shock becomes permanent. If a crisis-era decree (a Decreto de Necesidad y Urgencia, DNU, in Argentine practice) creates a new institutional arrangement, the IHR captures how much of that arrangement persists after the crisis ends. High IHR means institutions do not spring back; deformation is irreversible. The concept borrows from materials science, where hysteresis describes the lag between an applied force and its effect, but the application is to institutional dynamics.

2.3.2 Institutional Evolvability Index (IEI)

The IEI measures the system's capacity for adaptive change, decomposed into four components: Variation (V), the rate at which new institutional arrangements are proposed; Selection (S), the rate at which proposed arrangements are tested against reality; Competition (C), the intensity of competition between alternative arrangements; and Migration (M), the flow of institutional innovations across jurisdictional boundaries. The IEI, together with CLI and IHR, forms a diagnostic triangle: high CLI and high IHR with low IEI describes a system trapped in a suboptimal equilibrium.

2.3.3 Heteronomous Bayesian Updating (HBU)

HBU is the most theoretically novel component of the framework. Standard Bayesian updating assumes that agents update their beliefs about a norm's validity by observing whether the norm produces good outcomes. HBU posits that legal agents update their beliefs by observing whether *others are sanctioned* for non-compliance, not by observing whether the norm itself is effective. The updating rule is:

$$P(\text{norm valid} \mid \text{sanction observed}) = P(\text{sanction} \mid \text{valid}) \times P(\text{valid}) / P(\text{sanction})$$

The distinction matters because it generates a pathological learning loop. In high-CLI systems like Argentina, sanctions for non-compliance are highly visible: courts issue injunctions, regulators impose fines, professional associations censure deviants. All agents observe these sanctions and update their beliefs upward: the norm must be valid, because violators are punished. The result is increased compliance, increased perception of validity, and increased CLI. The system reinforces itself through a mechanism that is entirely independent of whether the norm actually produces good outcomes for anyone. HBU emerged as an unanticipated finding during the development of the simulation; I did not set out to model it. The agents' behaviour under high sanction visibility

produced convergence patterns that could not be explained by standard Bayesian updating, leading to the formalization of the heteronomous variant. This kind of internal modification of the theoretical framework under pressure from data is, again, one of Bunge's requirements for genuine science.

3. Simulation Architecture

3.1 From OASIS to Legal Dynamics

The simulation engine adapts OASIS (Open Agent Social Interaction Simulations), a scalable multi-agent framework developed by CAMEL-AI and collaborating institutions. OASIS was designed to simulate social media dynamics: agents with individual personalities interact on platforms resembling X or Reddit, producing emergent phenomena like information spreading, group polarization, and herd behaviour. The adaptation to legal-institutional dynamics required replacing every domain-specific component while preserving the modular architecture.

Table 2: Architectural Mapping from OASIS to Legal Simulation

OASIS Component	Legal Adaptation	Function
Agent Module	7 legal agent types	Role-differentiated institutional actors
Environment Server	LegalEnvironment	Norm registry, precedent database, coalitions
Recommendation System	IncentiveSystem	Sanction/reward structure driving behaviour
Time Engine	InstitutionalTimeEngine	Legislative cycles, judicial lag, reform windows
Action Space (23 social)	9 Legal Actions	CUMPLIR through REVERSAR
Emergent metrics	CLI, IHR, IEI	Institutional rigidity, hysteresis, evolvability

3.2 Agent Types and Parameters

The simulation populates the institutional environment with seven agent types, each with role-specific parameters calibrated against observable characteristics of real institutional actors. I emphasize that calibration means informed estimation from documented institutional features, not statistical estimation from a training sample. The distinction matters for interpreting the validation results in Section 4.

Table 3: Agent Types and Key Parameters (Argentina Scenario)

Agent Type	N	CRI	Inst. Capital	Key Feature
Judge (CSJN)	5	0.78	0.90	Veto power, long doctrinal memory

Legislator (oficialismo)	130	0.30	0.40	Party discipline, reform mandate
Legislator (opposition)	127	0.60	0.50	Blocking incentive
Union	3	0.85	0.80	Coalition capacity 0.95, mass mobilization
Regulated firm	50	0.20	0.30	Capture capacity, EVADIR option
Citizen	500	0.15	0.10	Low information, herd susceptibility
Regulator	1	0.40	0.60	Sanction authority, capture vulnerability

The total for the Argentina scenario is 816 agents. A single simulation round processes all agents in approximately 40 milliseconds; 100 rounds complete in under 4 seconds on commodity hardware. This computational efficiency is critical for Monte Carlo validation, which requires hundreds of independent replications.

The Coefficient of Resistance to Institutional Change (CRI) is the parameter that most directly addresses the conformity bias documented in OASIS. Yang et al. (2024) reported that LLM-based agents in OASIS are more susceptible to herd behaviour than humans. For institutional simulation, where resistance to change is the central phenomenon, this bias must be corrected. The CRI implements an asymmetric cost function:

$$cost_abandon = base_cost \times institutional_capital \times (1 + n_supporters)$$

$$cost_maintain = base_cost \times (1 - doctrine_fitness)$$

An agent maintains its current doctrinal position when $cost_abandon$ exceeds $cost_maintain$ adjusted by the CRI. A CSJN judge with institutional capital of 0.90 and four colleagues supporting the same doctrine faces an abandonment cost that dwarfs the maintenance cost even when the doctrine has low fitness (0.10). The model thus produces the doctrinal inertia observable in Argentine Supreme Court jurisprudence over the last three decades.

3.3 Legal Action Space

The simulation defines nine legal actions, replacing the 23 social actions of OASIS. Each action has a base cost, a set of eligible agent types, and an effect on the system state (norm fitness, CLI components, coalition structure).

Table 4: Legal Action Space

Action	Eligible Agents	Cost	System Effect
CUMPLIR	All	Low	Reinforces active norm
EVADIR	Citizen, Firm	Medium (sanction risk)	Signals norm inefficacy

LITIGAR	Firm, Citizen, Union	High	Generates precedent, mutates doctrine
IMPUGNAR_CN	Any litigant	Very high	Activates constitutional review
REFORMAR	Legislator, Executive	Political capital	Mutates norm, generates resistance
COALICIONAR	Union, Party, Lobby	Coordination cost	Amplifies blocking/support power
CAPTURAR	Firm, Interest group	Resources	Institutional deviation (PSO)
SANCIONAR	Judge, Regulator	Institutional legitimacy	HBU signal; enforcement
REVERSAR	Judge, Successor govt	Political + legal capital	Undoes reform, generates hysteresis

3.4 Simulation Engine

Each simulation round executes ten steps in sequence: (1) advance all agents (age, update capital); (2) decay norm fitness; (3) check reform trigger; (4) construct observation for each agent (current CLI, active norms, recent sanctions); (5) collect actions from all agents via `decide_action()`; (6) process actions (resolve outcomes based on probabilistic success functions); (7) resolve reform outcome if one is active; (8) propagate HBU cascade from sanctions emitted during the round; (9) update CLI components based on aggregate actions; (10) compute period metrics (CLI, IHR, IEI, mean fitness).

The emergent CLI computation deserves specific attention. In calibrated mode, the CLI starts at the target value from the scenario configuration and is modified incrementally by agent actions. In emergent mode, the CLI starts at 0.50 (neutral) and evolves exclusively from agent interactions. Each component is computed from observed action rates: `VetoPlayers` from the frequency of successful blocking actions, `JudicialReview` from the ratio of successful constitutional challenges to total reforms attempted, `AmendmentDifficulty` from the inverse of `REFORMAR` success rate, and `InformalNorms` from the divergence between formal norm vigency and observed compliance behaviour. An exponential moving average with smoothing factor 0.15 prevents oscillation.

3.5 Replication Package

Reproducibility is a prerequisite for any programme that claims scientific status. A standalone replication package is available at github.com/adrianlerer/816-agentes-institucionales-argentinos-EPT, designed so that any researcher, reviewer, or interested reader can replicate the Argentina simulation in under five minutes.

The package requires no API keys, no external services, and no dependencies beyond `numpy`, `pyyaml`, `matplotlib`, and `pytest`. A single command (`python run.py`) executes five independent Monte Carlo replications of 100 rounds each with 816 agents, producing the CLI time series, action distributions, and validation summary reported in Section 4.

The replication package is intentionally self-contained: it includes all source code, agent parameters, scenario configuration, historical data, validation scripts, and documentation needed to reproduce the results reported in Section 4, without depending on any external repository or service. The package includes the 23 Argentine historical reforms dataset, the complete agent parameter specifications, and documentation explaining the methodology, expected results, and frequently asked questions. All code is Python 3.11+ with no asynchronous operations or external API calls.

4. Empirical Validation

4.1 Monte Carlo Validation: Argentina

The primary validation tests the Argentina scenario against the documented outcome of labour flexibilization reforms during the Menem administration (1990s). The scenario configures 816 agents as described in Table 3, with a reform trigger at round 5 and 100 total rounds per replication. Five independent seeds produce the following results:

Table 5: Monte Carlo Validation (Argentina, 5 seeds)

Seed	Final CLI	Block Rate	Union COAL+LIT
42	0.9233	93.8%	65.0%
179	0.9235	93.8%	65.0%
316	0.9235	93.8%	65.0%
453	0.9235	93.8%	65.0%
590	0.9235	93.8%	65.0%

Three validation criteria are met. First, emergent CLI falls within the target range [0.80, 0.95]: mean 0.9235, error from target (0.89) of 0.033, below the 0.05 threshold. Second, reform block rate exceeds 70%: observed 93.8%. Third, union strategy is dominated by COALICIONAR plus LITIGAR at 65%, consistent with the historical record of CGT and CTA behaviour during labour reform episodes.

The low variance across seeds (CLI standard deviation 0.0001) indicates that the Argentina scenario produces a strong attractor. This is itself a finding: high-CLI systems converge to stable equilibria with little sensitivity to initial conditions. The institutional lock-in is structural, not stochastic.

4.2 Historical Cross-Validation (23 Argentine Reforms)

The companion dataset documents 23 Argentine labour reforms spanning 1974 to 2024, coded by government, legal instrument, legislative and judicial outcomes, union response, implementation status, and final outcome. Era-level analysis reveals a pattern consistent with the simulation:

Table 6: Reform Outcomes by Political Era

Era	N Reforms	Success Rate	Dominant Mechanism
Military (1976-1983)	3	0%	Judicial reversal post-transition
Menem (1989-1999)	6	33%	Union litigation + constitutional challenge
Kirchner (2003-2015)	4	0%	Pre-emptive coalition blocking
Macri (2015-2019)	4	25%	DNU reversal + judicial injunction
Milei (2024-)	3	33%	Constitutional challenge (ongoing)

The overall success rate of 22% (5 of 23) is consistent with the simulation's 93.8% blocking rate in adversarial scenarios. The discrepancy is partially explained by the simulation's focus on adversarial reforms (those facing organized resistance), while the historical dataset includes some consensual reforms that passed without serious opposition. The simulation does not currently distinguish between these two reform types; this is a limitation discussed in Section 5.

4.3 Cross-Country Leave-One-Out Validation

The 60-case dataset spans four jurisdictions: Argentina (24 cases), Chile (13), Brazil (12), and Spain (11). Leave-one-out cross-validation (LOO-CV) uses the CLI as a static predictor: for each case, the CLI of the relevant jurisdiction predicts whether the reform will be blocked ($CLI > 0.50$) or succeed ($CLI < 0.50$).

Table 7: LOO Cross-Validation Results by Country

Country	N	Accuracy	Baseline	Delta
Argentina	24	62.5%	45.8%	+16.7 pp
Chile	13	53.8%	38.5%	+15.3 pp
Brazil	12	41.7%	41.7%	+0.0 pp
Spain	11	54.5%	45.5%	+9.0 pp
Overall	60	56.7%	43.3%	+13.4 pp

The overall accuracy of 56.7% (95% CI: 44.1-68.4%) exceeds the baseline of 43.3% by 13.4 percentage points. This is a modest result that I present as a positive signal, not a triumph. The pattern across countries is informative: Argentina (62.5%) shows the strongest predictive power,

consistent with the hypothesis that high-CLI systems are more predictable because structural constraints dominate transient political dynamics. Chile and Spain show moderate improvement over baseline. Brazil performs at chance, likely reflecting the heterogeneity introduced by its federal structure, where national CLI may not capture state-level variation.

The 56.7% figure requires honest interpretation. A static CLI predictor does not capture temporal dynamics, coalition shifts, or crisis-driven reform windows. The simulation engine, which models these dynamics explicitly, should outperform the static predictor. This hypothesis is testable but requires reform-specific scenario configuration for all 60 cases, a task reserved for future work.

4.4 Emergent CLI Validation

The emergent CLI mode, where the index evolves from agent interactions without target injection, was validated with 20 replications of 100 rounds each for all four countries.

Table 8: Emergent vs. Calibrated CLI (20 runs, 100 rounds)

Country	CLI Target	Emergent Mean	Error	Pass Rate
Argentina	0.89	0.92	0.033	100%
Spain	0.51	0.48	0.030	100%
Brazil	0.40	0.38	0.020	100%
Chile	0.24	0.22	0.020	100%

All four countries fall within the acceptance criterion of $|\text{emergent} - \text{target}| < 0.10$, with a 100% pass rate across 80 total replications (20 per country). Crucially, the ordinal ranking is preserved: Argentina > Spain > Brazil > Chile. This means that the relative rigidity of these institutional systems emerges from agent interactions alone, without being imposed by calibration. The CLI, in this sense, is not merely a descriptive label applied to known quantities; it is a property that the simulation derives from institutional structure.

I note that the emergent values consistently overshoot for high-CLI systems (Argentina: 0.92 vs. 0.89) and undershoot for low-CLI systems (Chile: 0.22 vs. 0.24). This systematic bias, approximately 0.03 in magnitude, may reflect the asymmetric cost function in the CRI: the resistance mechanism is slightly stronger than the empirical calibration suggests. Alternatively, the target CLI values themselves may contain estimation error of similar magnitude. Both possibilities are consistent with the data; I cannot distinguish between them with the current design.

5. Discussion

5.1 Biology as the Right Analogy

The Huang parallel is not merely rhetorical. It identifies a structural transition that has occurred in multiple domains: physics became engineering when Newton's laws enabled prediction; chemistry became engineering when the periodic table enabled systematic molecular design; biology is becoming engineering as computational protein folding enables drug design by simulation. In each case, the transition occurs when three conditions are met: (a) the phenomena are formalized in computable models, (b) simulation becomes cheaper than experimentation, and (c) each simulation result improves the model.

The EPT framework proposed here attempts to establish condition (a) for legal-institutional dynamics. The simulation engine provides condition (b): 100 rounds of institutional simulation with 816 agents in under 4 seconds. Condition (c), the compounding of knowledge across simulations, requires infrastructure that does not yet exist for law but has been proposed under the Criptolus framework, where JurisRank measures precedent fitness and accumulated precedents train the system's predictive capacity.

The analogy also identifies the key difference between biological and legal engineering. Molecules do not make strategic decisions; institutional agents do. Goodhart's Law ('when a measure becomes a target, it ceases to be a good measure') applies to legal reform but not to protein folding. This means that legal-institutional simulation faces an adversarial dimension absent from computational biology: the agents being simulated may alter their behaviour in response to the simulation's predictions. This is not a fatal objection; it is a design constraint that the simulation must account for, analogous to how economic models account for rational expectations.

5.2 Limitations

Static CLI components. The current model treats the constitutional structure as fixed within a simulation run. In reality, constitutional amendments occur (rarely in Argentina, more frequently in Chile and Brazil). A fully dynamic model would allow the genotype itself to mutate, not just the phenotypic expression. This is architecturally feasible but not yet implemented.

Binary reform scenarios. The simulation models reforms as adversarial (facing organized resistance) or successful (passing without serious opposition). Real reforms exist on a spectrum. The 56.7% LOO accuracy likely reflects this oversimplification: reforms that were partially implemented or implemented with significant modifications are difficult to classify as successes or failures.

No LLM-driven agents. The current agents are rule-based, with decision functions defined by the CRI, HBU, and cost parameters. Integrating LLM-driven agents (as in OASIS) would enable richer reasoning but would also introduce the conformity bias that the CRI was designed to correct. The sequence matters: first validate the rule-based engine, then layer LLM reasoning on top.

Small dataset. Sixty cases across four countries is a small sample by any standard. The 95% confidence interval for overall accuracy spans 24.3 percentage points (44.1-68.4%). Expansion to additional jurisdictions and reform types would narrow this interval and improve the programme's empirical foundation.

Bunge's game theory objection. The programme uses evolutionary game theory extensively. Bunge considered many game-theoretic applications pseudoscientific because utilities are subjective and manipulable. The fitness metrics proposed here (adoption rate / litigation rate) are more observable than subjective utilities, but the cost functions driving agent decisions still contain estimated parameters. This objection is not resolved; it is mitigated.

5.3 Policy Implications

Two implications follow from the validation results. First, high-CLI systems like Argentina require coalition-based reform strategies. The simulation shows that unilateral executive action (DNU-based reform) is blocked at rates exceeding 90% when unions activate COALICIONAR plus LITIGAR. Successful reform in high-CLI environments requires either reducing the number of effective veto players (constitutional reform, which is itself blocked by high CLI) or building coalitions that include potential blockers.

Second, the existence of validated simulation tools changes the standard of care (*lex artis*) for legislative design. Under Argentine law (Articles 1757-1758, Civil and Commercial Code), activities that create risk generate strict liability. If simulation tools that can predict institutional outcomes are available and a legislature enacts a reform without using them, the omission may constitute a breach of the duty of care. This argument was developed formally in a companion paper on pre-deployment normative evaluation, proposing a three-phase protocol analogous to clinical trials: design, simulation with synthetic populations, and controlled pilot.

6. Conclusion

This paper has attempted to do three things. First, to take seriously the proposition that law can transition from a science of sporadic discovery to an engineering discipline of compounding improvement. Second, to build and validate a simulation tool that demonstrates this transition is feasible with current technology. Third, to submit the entire programme to the demarcation criteria of Mario Bunge, documenting both where it passes and where it falls short.

The results are encouraging but incomplete. The CLI emerges from agent interactions without target injection, preserving the ordinal ranking of institutional rigidity across four countries. Monte Carlo validation against Argentine historical data produces reform-blocking rates consistent with observed outcomes. Cross-country validation shows modest but above-baseline predictive power. These findings suggest that institutional dynamics are, at least partially, computable.

The programme's weaknesses are real. Sixty cases is not enough. The utility functions in the game-theoretic components are not fully observable. The simulation does not yet distinguish adversarial from consensual reforms. The emergent CLI systematically overshoots for rigid systems and undershoots for flexible ones. These are not concessions appended to satisfy reviewers; they are the next items on the research agenda.

What this paper represents, however, is not a conclusion but a starting point. The pharmaceutical industry did not arrive at modern drug regulation overnight. It took decades to move from unregulated therapeutic experimentation on human subjects to the structured sequence of preclinical testing, Phase I-III clinical trials, and post-market surveillance that defines the current standard of care. The path was neither linear nor uncontested; each phase was built on the failures and lessons of the previous one.

The research programme described here proposes an analogous path for normative design. The 816-agent simulation is a first preclinical trial: a synthetic population subjected to a normative intervention under controlled conditions, with measurable endpoints and documented failure modes. The next step is not to scale the number of agents or jurisdictions, although that will happen. The next step is to build the institutional infrastructure for systematic pre-deployment normative evaluation: what might be described, in the Argentine regulatory tradition, as an ANMAT for legislation¹.

An ANMAT for legislation would not mean that every bill requires computational simulation before reaching the floor of Congress. It would mean that norms with foreseeable massive impact, those that affect millions of people, restructure labour markets, modify fiscal regimes, or alter constitutional equilibria, are subjected to a structured evaluation protocol before deployment. Phase 1: textual analysis of systemic coherence and constitutional consistency. Phase 2: simulation with synthetic populations calibrated against real institutional data, measuring expected blocking rates, litigation cascades, and fiscal costs of judicialization. Phase 3: geographically or temporally limited pilot implementation with predefined success and failure metrics. The tools for Phase 2 now exist. This paper demonstrates that they produce results consistent with historical data.

Jensen Huang observed that when a field becomes engineering, improvement becomes exponential. Each validated simulation refines the model. Each jurisdiction added to the dataset improves cross-validation. Each reform predicted and then observed, whether the prediction succeeds or fails, trains the next iteration. The replication package at github.com/adrianlerer/816-agentes-institucionales-argentinos-EPT allows anyone to verify, replicate, and extend the claims in this paper.¹⁹ That openness is not incidental; it is constitutive. A programme that claims scientific status under Bunge's criteria must expose its methods and data to scrutiny. This one does.

¹ LERER, I. A. (2026). Pre-Deployment Normative Evaluation: An Extended Phenotype Framework for Legislative Quality Assurance (1.0). Zenodo. <https://doi.org/10.5281/zenodo.18947186>

The question that remains is not technical but institutional. The pharmaceutical industry adopted clinical trials because the cost of not doing so, measured in human suffering and corporate liability, became intolerable. The cost of legislative trial-and-error, measured in failed reforms, wasted political capital, and populations trapped in suboptimal institutional equilibria, has not yet reached the threshold that forces the adoption of new methods. This paper argues that the threshold should be recalibrated. The tools exist. The standard of care has changed. What remains is for the legal community to act accordingly.

Replication package, Public GitHub Repository: <https://github.com/adrianlerer/816-agentes-institucionales-argentinos-EPT>

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